

Domain Name System (DNS) notes

The Domain Name System maps domain names (www.google.com) to information such as IP-addresses (142.251.142.238).

The service works on the application layer.

The use case is to have human-memorable addresses instead of generic IP-addresses.

There are several DNS servers at different network layers.

- The local computer may have a local resolver and cache. On my Ubuntu system, systemd-resolved listens at 127.0.0.53 .
- The router may forward DNS queries from devices on the home network and may cache answers.
- The Internet Service Provider has one.
- Authoritative DNS servers hold the DNS records for particular domains and provide answers to resolvers.

Name resolution note:

Entering the address 'http://example.com/page' does not send the whole string to the DNS server. Only 'example.com' needs resolution. The rest ('/page') is used as a http request.

Hierarchical structure - Domains and Zones.

```
+-----+      -
|root domain|  : root domain
|   .   |     :
+-----+      -
|   |   |   |  : Top level domains
+----+ |   |   |  :
|.com| |   |   |  :
+----+ |   | +----+ :
      |   | |.net| :
      +----+ | +----+ :
      |.org| |   | :
      +----+ +----+ :
      |   | |.se | :
      |   | +----+ -
      |   |       : ZONE CNAME
```

```
+-----+ :  
|en.wikipedia.org| :  
+-----+ -
```

```
www.hkr.se  
^   ^   ^- Top level domain  
|   ^- Domain name  
^- Computer name - HTTP server in this case.
```

Thirteen root servers spread across the world.
They keep the top level domains (.com, .net, .org ...)

Recursive question.

User runs: `resolvectl query www.hkr.se`

1. `resolvectl` asks the local `systemd-resolved` service to resolve the name.
2. `systemd-resolved` checks whether it already has an answer cached.
3. If it needs an external answer, it sends DNS queries for address records (A for IPv4 and AAAA for IPv6) to its configured upstream DNS server.
4. On your previously shown setup, that upstream server was your router, 192.168.0.1. The router may answer from its cache or forward the question to another resolver, such as one operated by your ISP.
5. The recursive resolver obtains an answer, potentially consulting the DNS hierarchy, and returns it along the chain.
6. `systemd-resolved` returns the address to `resolvectl`.

Iterative question

1. Can you give me the address to 'www.hkr.se'?
2. Possible answers:
 - yes
 - no
 - Check with ...

Iterative resolution of an A record for www.hkr.se

1. Resolver → root server:
"What is the A record for www.hkr.se?"
Root → resolver:
"Ask the name servers for .se." [referral]

2. Resolver → .se name server:
"What is the A record for www.hkr.se?"
.se → resolver:
"Ask the name servers for hkr.se." [referral]
3. Resolver → hkr.se authoritative server:
"What is the A record for www.hkr.se?"
Server → resolver:
"Here is the answer." [answer]

The record may be cached, so it maybe doesnt need to ask the question.

Caching in DNS servers